السلام عليكم ورحمة الله وبركاته.. وبعد،

بين يديك ملخص شامل ومنظم للمادة، تم إعداده بعناية ليكون عوناً لك في فهم المادة .وقبل البدء، أضع بين يديك ميثاقاً للأمانة العلمية والتعاون:

⚠️ ميثاق الأمانة والمسؤولية:

هذا العمل هو جهد طالب، أضعه بين يديك بنية النفع والفائدة؛ لذا:

- لا أسامح ولا أحلل استخدامه في الغش أو أي عمل ينافي الأخلاق الإسلامية.
- يُمنع منعاً باتاً تداوله كمصدر تجاري للباعة أو استخدامه في أدوات الذكاء الاصطناعي لغرض التحايل او الغش.
- الأمانة العلمية تقتضي منك استخدامه كمرجع للفهم والمراجعة، لا كبديل عن الضمير والاجتهاد الشخصي.

📣 التواصل معي:

الكمال لله وحده، لذا يسعدني استقبال ملاحظاتكم، تعديلاتكم، أو إضافاتكم عبر:

- الرقم المباشر: 0530717788
- الموقع الشامل: kau-one.web.app

توجيه إيماني:

قبل أن تبدأ، استعذ بالله من الشيطان الرجيم، وجدد نيتك بأن يكون هذا الوقت في سبيل طلب العلم النافع الذي يرفع قدرك في الدنيا والآخرة. قال رسول الله ﷺ: "مَنْ سَلَكَ طَرِيقًا يَلْتَمِسُ فِيهِ عِلْمًا سَهَّلَ اللَّهُ لَهُ بِهِ طَرِيقًا إِلَى الْجَنَّةِ" (رواه مسلم).

إعداد الطالب: إلياس حسان بابور الدفعة: 24 — تم اخذ الماده و عمل الملخص على السلايدات سنه م2026

1. What is the Internet? The slides defines the Internet from two main perspectives:

A. The "Nuts and Bolts" View (منظور البنية التحتية والمكونات المادية):

- **Hosts / End Systems:** Millions of connected computing devices running network applications. These include traditional devices (PCs, servers, smartphones) and non-traditional appliances (smart TVs, IP picture frames).
- **Communication Links:** The physical media connecting the network, such as fiber optic, copper, radio, and satellite. The transmission rate of these links is called **bandwidth**.
- **Packet Switches:** Devices that forward chunks of data (called **packets**) through the network. The two main types are **routers** and **switches**.
- **Network of Networks:** The Internet is formed by interconnecting multiple ISPs (Internet Service Providers - مزودي خدمة الإنترنت) globally and regionally.

B. The Service View (منظور الخدمات):

- **Infrastructure for Applications:** It acts as the foundational infrastructure that provides services to distributed applications (like Web, VoIP, email, e-commerce, and games).
- **Programming Interface:** It provides "hooks" (interfaces) for applications to connect to the Internet and send/receive data, offering different service options similar to how a postal system works.

2. Protocols and Standards (البروتوكولات والمعايير)

A. What is a Protocol? Just like humans have social rules for communication (e.g., saying "Hi" before asking a question), computer networks rely on protocols. All communication activity in the Internet is governed by them.

- **Definition:** A protocol defines the **format** and **order** of messages sent and received among network entities, and the **actions taken** upon message transmission or receipt.
- **Examples:** TCP, IP, HTTP.

B. Internet Standards (معايير الإنترنت): To ensure all devices communicate smoothly, standards are enforced by specific organizations:

- **IETF (Internet Engineering Task Force):** The body responsible for developing Internet standards.
- **RFC (Request for Comments):** The official documents published by the IETF that define how specific protocols work.
- **IEEE 802:** The committee responsible for local area network (LAN) and wireless standards (e.g., Ethernet, Wi-Fi 802.11).

1. Network Structure Overview The network is divided into three main components:

- **Network Edge:** Where the end systems (hosts, clients, and servers) reside. Servers are often located in large data centers.
- **Access Networks & Physical Media:** The wired and wireless communication links that connect the edge to the rest of the network.
- **Network Core:** The interconnected routers that form the "network of networks."

2. Access Networks These networks connect end systems to the first edge router. When evaluating them, two key factors matter: **Bandwidth** (speed in bits per second) and whether the link is **Shared or Dedicated**.

- **Digital Subscriber Line (DSL):**
  - Uses existing dedicated telephone lines connected to a DSLAM at the central office.
  - Data and voice are transmitted at different frequencies over the same line (data goes to the Internet, voice to the telephone network).
  - Speeds are asymmetrical: Downstream (< 24 Mbps) is much faster than Upstream (< 2.5 Mbps).
  - خط الاشتراك الرقمي *(DSL):* يستخدم خطوط الهاتف النحاسية الحالية. يتم فصل البيانات عن الصوت باستخدام ترددات مختلفة، وعادة ما تكون سرعة التنزيل أعلى بكثير من سرعة الرفع.
- **Fiber to the Home (FTTH):**
  - Provides a direct optical fiber path from the central office to the home.
  - Extremely fast (up to Gigabits/second), making it superior to DSL and traditional cable.
- **Home Networks:** Typically consist of a single box combining a cable/DSL modem, a router/firewall/NAT, wired Ethernet, and a wireless access point.
- **Enterprise Access Networks (Ethernet):**
  - Used in institutions (companies, universities).
  - End systems connect to an Ethernet switch, which connects to an institutional router. Offers extremely high transmission rates (10 Mbps up to 10 Gbps).
- **Wireless Access Networks:** Connect end systems to a router via a base station (access point).
  - *Wireless LANs (WiFi):* Local range (within a building / ~100 ft).
  - *Wide-Area Wireless (Cellular):* Provided by telecom operators over tens of kilometers (e.g., 3G, 4G, LTE).

3. Physical Media The physical link carries bits between a transmitter and a receiver. They are categorized into two types:

A. Guided Media (Solid Media): Signals propagate through physical matter.

- **Twisted Pair (TP):** Two insulated copper wires (e.g., Cat 5 for up to 1 Gbps, Cat 6 for 10 Gbps).
- **Coaxial Cable :** Two concentric copper conductors. It is bidirectional and supports broadband (multiple channels on one cable).
- **Fiber Optic Cable (كابلات الألياف الضوئية):** Glass fibers carrying light pulses (each pulse is a bit). Very high speed, low error rate, and immune to electromagnetic noise. Repeaters can be spaced far apart.

B. Unguided Media (Radio): Signals propagate freely in the electromagnetic spectrum without physical wires.

- **Challenges:** Affected by the environment, including reflection, physical obstructions, and interference.

- **Types of Radio Links:**
    - *Terrestrial Microwave:* Up to 45 Mbps.
    - *LAN (WiFi):* Local wireless.
    - *Wide-Area (Cellular):* Few Mbps over large distances.
    - *Satellite (*الأقمار الصناعية*):* Kbps to 45 Mbps. Suffers from a high **270 msec end-to-end delay** (latency) due to the massive distance in space.

The Network core as a Mesh & Full Link Capacity

The network core is essentially a mesh of interconnected routers. In packet-switching, hosts

break down application-layer messages into smaller packets. These packets are then forwarded from one router to the next across the links along the path from the source to the destination. A crucial characteristic of packet switching is that each packet is transmitted at the full link capacity of that specific link.

1. Packet Switching

Hosts break application messages into smaller chunks called **packets** (length L bits) and transmit them into the access network at a transmission rate R (link capacity/bandwidth).

- **Transmission Delay:** The time needed to push a packet into the link is L/R seconds.
- **Store-and-forward:** A router must receive the *entire* packet before it can begin transmitting it to the next link.
- **Queueing Delay and Loss:** If packets arrive at a router faster than the link's transmission rate, they queue in a buffer. If the buffer memory fills up completely, newly arriving packets are dropped (lost).

2. Two Key Network-Core Functions

- **Routing:** Determines the complete end-to-end route a packet takes from source to destination using routing algorithms.
- **Forwarding:** The local action of moving a packet from a router's input port to the appropriate output port based on a forwarding table.

3. Circuit Switching

Resources are explicitly allocated and reserved for a specific "call" between source and destination (common in traditional telephone networks).

- **Dedicated Resources:** No sharing with other users, which guarantees circuit-like performance.
- **Idle Segments:** If the reserved circuit is not actively being used by the call, the bandwidth is wasted.
- **Multiplexing Types:** Uses **FDM** (Frequency Division Multiplexing - dividing the spectrum into frequency bands) or **TDM** (Time Division Multiplexing - dividing time into slots).

4. Packet Switching vs. Circuit Switching

Packet switching allows significantly more users to share a network. For example, on a 1 Mbps link where users need 100 kb/s but are only active 10% of the time, circuit switching supports exactly 10 users. Packet switching can support 35 users with a nearly zero probability of congestion.

- **Packet Switching Pros:** Excellent for bursty data, allows resource sharing, and requires no call setup.
- **Packet Switching Cons:** Can lead to excessive congestion, causing packet delay and loss. It requires complex protocols for reliable data transfer and congestion control.

- تحويل الحزم *(Packet Switching)* يسمح لعدد أكبر بكثير من المستخدمين بمشاركة الشبكة لأنه يستغل فترات خمول المستخدمين. ممتاز للبيانات المتقطعة ولا يحتاج لإعداد اتصال مسبق.
- عيوبه: قد يسبب ازدحاماً يؤدي إلى تأخير وفقدان الحزم، مما يتطلب بروتوكولات للتحكم في الازدحام.

5. Internet Structure: Network of Networks

Connecting millions of access ISPs directly to each other is impossible (it would require too many direct connections).

- **Tier-1 ISPs:** Global transit providers (e.g., Sprint, AT&T). Access ISPs connect to these global ISPs.
- **IXP (Internet Exchange Point):** Competitor global ISPs connect to each other via IXPs and peering links to ensure data can reach anywhere.
- **Regional Networks:** Arise to connect smaller local access networks to the large global ISPs.
- **Content Provider Networks:** Companies like Google or Microsoft run their own massive private networks. They place data centers close to end-users, often bypassing Tier-1 and regional ISPs entirely to improve speed and reduce costs.

next is 1.4 in chapter 1

How do loss and delay occur?

packets queue in router buffers

packet arrival rate to link (temporarily) exceeds output link capacity packets queue, wait for turn - packet being transmitted (delay) - **packets** queueing (delay) - free (available) buffers: arriving packets dropped (loss) if no free buffers

The 4 Sources of Delay

When a packet travels from your computer (A) to a server (B), it stops at several routers along the way. At *each* router, it experiences four different types of delay.

Detailed Breakdown:

Total delay at a single node (router) is the sum of these four:
$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

1. **Processing Delay (d_{proc}):** The router examines the packet to see where it needs to go next and checks for bit errors. (Very fast, usually < 1 millisecond).
2. **Queueing Delay (d_{queue}):** The packet waits in a line (buffer) because the router is busy sending other packets. (Variable, depends on network congestion).
3. **Transmission Delay (d_{trans}):** The time it takes to push all the bits of the packet onto the physical wire.
    - Formula: L/R (L = Packet length in bits, R = Link bandwidth/speed in bps).
4. **Propagation Delay (d_{prop}):** The time it takes for the bit to physically travel through the wire/fiber to the next router.
    - Formula: d/s (d = Distance of the link, s = Speed of light/signal in the medium).

أي رسالة ترسلها تمر براوترات. في كل راوتر يصير لها 4 أنواع من التأخير:

1. **المعالجة (Processing):** الراوتر يفحص الرسالة ويقرر وين يرسلها.
2. **الانتظار (Queueing):** الرسالة توقف في طابور إذا كان الراوتر مشغول.
3. **وقت الدفع (Transmission):** الوقت اللي يحتاجه الراوتر عشان "يدف" أو يضخ البيانات في السلك. يعتمد على سرعة النت وحجم الملف.
4. **وقت السفر (Propagation):** الوقت اللي تحتاجه الإشارة عشان تسافر في السلك من مدينة لمدينة. يعتمد على المسافة.

Real-World Example (The Airport):

- **Processing:** Checking your passport at the gate.
- **Queueing:** Waiting in line in the tunnel to get on the plane.
- **Transmission:** The time it takes for all 200 passengers to walk through the plane door.
- **Propagation:** The actual flight time in the air from Riyadh to Dubai.

Transmission vs. Propagation - The Caravan Analogy

People often confuse Transmission and Propagation. The slides use a "Caravan of Cars" analogy to clear this up.

Detailed Breakdown:

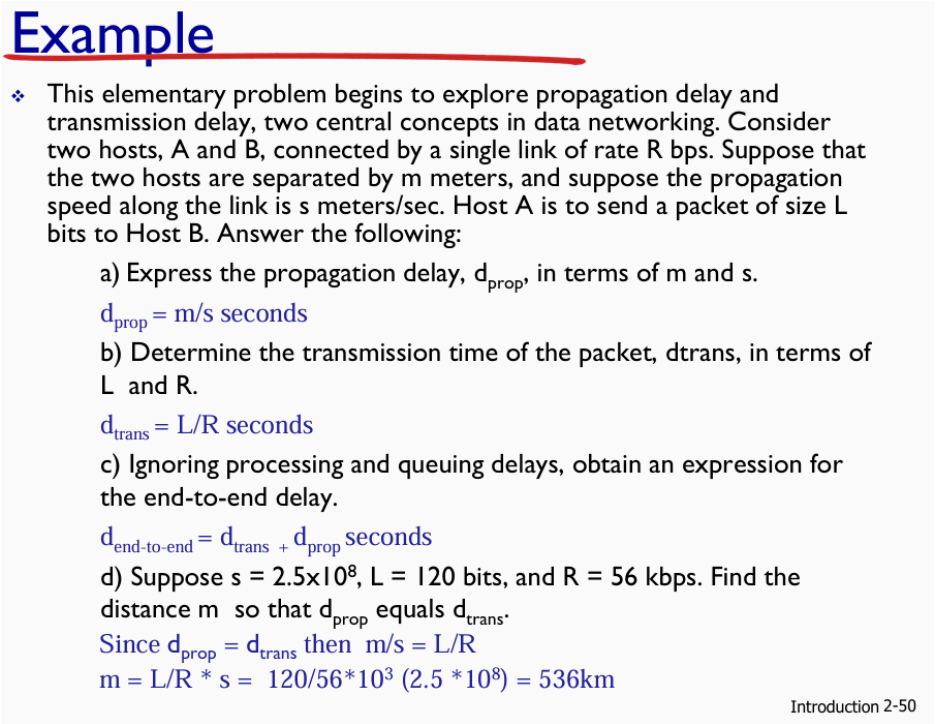
- Imagine a 10-car caravan (10 cars = 1 packet).
- **Transmission time** is the Toll Booth. It takes 12 seconds to service one car. To get all 10 cars through the booth, it takes 120 seconds (2 minutes). *This is pushing the packet onto the link.*
- **Propagation time** is driving on the highway. The cars drive at 100 km/hr to the next booth 100 km away. It takes 1 hour for a car to make that trip. *This is the packet traveling through the wire.*

Queueing Delay & Packet Loss (فقدان البيانات)

What causes lag and disconnected calls? The queue (the waiting room inside the router).

Detailed Breakdown:

- **Traffic Intensity (La/R):** This formula checks how busy the router is.
  - If La/R is close to 0, there is no line. Delay is small.
  - If La/R is close to 1, the line is getting huge. Delay is large.
  - If $La/R > 1$, data is arriving faster than the router can send it out. Delay becomes infinite!
- **Packet Loss:** Routers have limited memory (buffers). If a packet arrives and the queue is 100% full, the router simply throws the packet away. It is "dropped" or "lost."



Traceroute & End-to-End Delay

How do we know how long it takes to reach a destination? We use a program called traceroute.

Detailed Breakdown:

- **Traceroute:** A program that sends 3 test packets to *every single router* between you and the destination. It measures how many milliseconds it takes to get a response back.
- **End-to-End Delay:** If you assume there is no traffic jam (no queueing delay), the total time it takes for a packet to reach its destination across N routers is: $d_{end-end} = N(d_{proc} + d_{trans} + d_{prop})$

Example:

Traceroute is like a tracking number on an Amazon package. It tells you exactly which cities your package stopped in, and how long it took to leave each city.

Throughput and Bottlenecks (الإنتاجية وعنق الزجاجة)

Throughput is the actual speed at which bits are transferred from sender to receiver. Your internet speed is only as fast as its slowest link.

Detailed Breakdown:

- Imagine a server sending you a file. The server has a link speed of R_s, and your home internet has a link speed of R_c.
- **The Bottleneck Link:** The actual speed of your download will be the *minimum* of (R_s, R_c, or the network core speed).
- If the server can send at 100 Mbps, but your home Wi-Fi is only 10 Mbps, your throughput is 10 Mbps. The home network is the bottleneck.

Example:

Imagine drinking a very thick milkshake. It doesn't matter how big your mouth is or how big the cup is; the speed you drink it depends entirely on how wide the straw is. The straw is the bottleneck link.

next is 1.5

Why Layering? (The Airline Analogy)

The Internet is massive (hardware, software, routers, links). If we tried to build it as one giant piece of software, it would be a nightmare. Instead, we use **layers**.

Detailed Breakdown:

- The slides use an **airline analogy**. When you fly, the process is divided into steps: Buying a ticket -> Checking baggage -> Boarding at the gate -> Takeoff.
- **Modularization:** Because it is layered, if the airport changes how baggage is handled, the pilots don't need to learn a new way to fly the plane.
- In networks, each layer provides a specific service to the layer directly above it. If we invent a new type of WiFi (Link layer), your web browser (Application layer) doesn't need an update to keep working.

The 5-Layer Internet Protocol Stack

This is the actual blueprint of the Internet. Every time you send a message, it travels down these 5 layers on your device, crosses the internet, and travels up the 5 layers on your friend's device.

Detailed Breakdown:

1. **Application:** The apps you actually use (Browsers using HTTP, Email using SMTP).
2. **Transport:** Moves data directly between processes/apps. (TCP is **reliable**, UDP is **fast** but **doesn't check for errors**).
3. **Network:** Finds the best path to send data from the source computer to the destination computer across the globe (uses IP addresses and routing).
4. **Link:** Moves the data just one hop to the neighboring router or switch (e.g., Ethernet, WiFi).
5. **Physical:** The actual physical hardware—turning data into electrical signals or light (bits "on the wire").

Example:

Think of ordering a product online:

- **Application:** The Amazon app where you buy the item.
- **Transport:** The logistics manager who groups your items into a specific box.
- **Network:** The postal system deciding which flight/truck path is fastest to your country.
- **Link:** The specific delivery van driving from the warehouse to the airport.
- **Physical:** The actual roads and tires of the van.

The OSI Model vs. The Internet Stack

In the early days, a committee (ISO) created a 7-layer model called the OSI model. But the real Internet we use today only uses 5 layers.

Detailed Breakdown:

- The OSI model included two extra layers just below the Application layer:
    - **Presentation:** Formatting data, encryption, and compression.
    - **Session:** Keeping track of check-points in case a download gets interrupted.
- **Why is the Internet missing these?** The creators of the Internet decided they weren't strictly necessary for *every* application. If a modern app (like a bank app) needs encryption or sessions, the app developer just builds it directly into the Application layer.

next 1.6

Encapsulation

As your data moves down the 5 layers to be sent out, each layer adds its own "tag" or "header" with instructions.

Detailed Breakdown:

- You start with a **Message (**M**)**.
- The Transport layer takes the message and attaches a header (H_t). Now it's called a **Segment**.
- The Network layer takes that segment and adds its own header (H_n). Now it's called a **Datagram**.
- The Link layer adds its header (H_l). Now it's called a **Frame**.
- When it reaches the destination, the receiving computer strips the headers off one by one (Decapsulation) until it reads the original message.

Network Security (The Bad Guys)

The Internet was originally built for a small group of trusted scientists, so security was an afterthought. Now, engineers have to play "catch-up" to defend against attacks.

Detailed Breakdown:

- **Malware:** Malicious software. Includes Spyware (stealing passwords/keystrokes) and Botnets (hijacking your computer to attack others).
    - **Virus vs. Worm:** A *Virus* needs you to do something stupid (like opening an infected email attachment) to run. A *Worm* is self-replicating and passive—it infects your computer automatically over the network without you doing anything.
- **Denial of Service (DoS):** Attackers flood a server with massive amounts of fake traffic so that real users can't access it.
- **Packet Sniffing:** If you are on public WiFi (broadcast media), a "sniffer" program can silently copy all the unencrypted data flying through the air, including passwords.
- **IP Spoofing:** An attacker modifies the packet so it has a fake "return address," making it look like it came from someone you trust.

last part in chapter 1 :

Era 1: 1961–1972 (The Birth of Packet Switching)

The Big Picture: Before this era, phone calls used "Circuit Switching" (reserving a dedicated line). Scientists proved that data should instead be chopped into pieces ("Packet Switching"). This era is the birth of the very first network.

Detailed Breakdown:

- **1961-1964:** Researchers like Kleinrock and Baran proved the theory of packet switching.
- **1969:** The US military/research agency created **ARPAnet**. This was the very first working version of the Internet.
- **1972:** ARPAnet had 15 nodes (computers) connected. The first email program was written, and the first protocol (NCP) was used.

الشرح بالعربي (Arabic Context): البدايات الحقيقية. قبل هذي الفترة، كانت الاتصالات تعتمد على خطوط الهاتف الثابتة (Circuit Switching). العلماء أثبتوا إن نقل البيانات أفضل لو قسمناها لحزم صغيرة (Packet Switching). وزارة الدفاع الأمريكية أسست أول شبكة فعلية اسمها **ARPAnet**، وفي هذي الفترة تم إرسال أول إيميل في التاريخ.

Era 2: 1972–1980 (Connecting the Networks)

The Big Picture: People started building *different* networks (like ALOHAnet in Hawaii and Ethernet at offices), but they couldn't talk to each other. We needed rules to link them together.

Detailed Breakdown:

- **Cerf and Kahn (1974):** These two engineers wrote the "rules" for internetworking (connecting networks together). Their principles still run the Internet today:
    - *Minimalism:* Don't force networks to change their internal hardware to connect.
    - *Best effort service:* The network will *try* its best to deliver data, but doesn't guarantee it (that's the end-device's job).
    - *Stateless routers:* Routers shouldn't keep complex records of every connection; they just pass packets along.
    - *Decentralized control:* No single computer is in charge of the whole internet.

الشرح بالعربي (Arabic Context): ظهرت شبكات كثيرة ومختلفة بس ما كانت تفهم بعض. العالمان سيرف وكان (Cerf and Kahn) حطوا القواعد الأساسية لربط هذي الشبكات ببعض (Internetworking). قواعدهم هي نفس القواعد اللي تمشي عليها الإنترنت اليوم: شبكة لامركزية (مافي جهة وحدة تتحكم بكل شيء)، وراوترات بسيطة وظيفتها بس تمرر البيانات (Stateless routers).

Era 3: 1980–1990 (The Protocol Boom)

The Big Picture: The network grew rapidly, and all the famous standard languages (protocols) we still use today were officially defined and deployed.

Detailed Breakdown:

- **1983 - TCP/IP:** This became the official universal language of the Internet.
- **DNS (Domain Name System):** Invented to translate human words (like `google.com`) into IP addresses (like `192.168.1.1`).
- **SMTP & FTP:** Standard protocols for Email and File Transfers were established.
- The network exploded to over 100,000 computers connected.

الشرح بالعربي (Arabic Context): مرحلة التوحيد والنمو. تم اعتماد **TCP/IP** كلغة رسمية للإنترنت. وتم اختراع الـ **DNS** (لأرقام صعبة). وأيضاً تم تخيل إنه دليل الهاتف حق الإنترنت، يحول اسم الموقع السهل لـ IP. تحديد قوانين الإيميل (SMTP). الشبكة كبرت وصار فيها مئات الآلاف من الأجهزة.

Era 4: 1990–2000s (The Web & Commercialization)

The Big Picture: The Internet went from a boring text-based tool for scientists to a colorful, commercial playground for the public.

Detailed Breakdown:

- **The World Wide Web:** Invented by Tim Berners-Lee (HTML, HTTP). Before this, the Internet was just raw data. The Web gave us clickable links and pages.
- **Commercialization:** The government lifted restrictions, allowing businesses to make money on the Internet.
- **Killer Apps:** Web browsers (Mosaic, Netscape), instant messaging, and Peer-to-Peer (P2P) file sharing arrived.
- Because so many normal people joined, **security** became a major concern for the first time.

Real-World Example: Think of the *Internet* as the physical roads and infrastructure. Think of the *Web (WWW)* as the shopping malls, billboards, and restaurants built *on* those roads. Tim Berners-Lee built the first mall.

الشرح بالعربي (Arabic Context): "الإنترنت للجميع! العالم تيم بيرنرز لي اخترع "الشبكة العنكبوتية (Web - HTML/HTTP). جاب لنا المواقع Web قبلها الإنترنت كان شاشات سوداء وأكواد، لكن الـ والروابط والصور. ظهرت المتصفحات، وبدأ الاستخدام التجاري للإنترنت، وظهرت برامج المحادثة. ومع زيادة الناس، بدأ الاهتمام بموضوع "أمن الشبكات" (Security).

Era 5: 2005–Present (Smartphones, Clouds, and Tech Giants)

The Big Picture: This is the Internet you know today. It's wireless, mobile, massively centralized in "the cloud," and heavily influenced by a few giant corporations.

Detailed Breakdown:

- **Smartphones & Broadband:** Everyone is connected 24/7 with high-speed WiFi and cellular data.
- **Social Networks:** The rise of Facebook, Twitter, etc., bringing a billion+ users online.
- **The Cloud:** Companies stopped buying their own servers and started renting server space from Amazon (AWS) or Microsoft (Azure).

- **Private Networks:** Tech giants like Google and Microsoft grew so big that they built their *own* private underwater cables and networks to bypass the public Internet, making their services faster.

الشرح بالعربي (Arabic Context): العصر الحديث. عصر الجوالات الذكية والواي فاي السريع في كل مكان. ظهرت وسائل التواصل الاجتماعي (زي فيسبوك). والأهم هو ظهور "الحوسبة السحابية (Cloud)؛ الشركات صارت تحط بياناتها في سيرفرات أمازون بدل ما تشتري سيرفرات خاصة. الشركات العملاقة زي جوجل صار عندها شبكات إنترنت خاصة فيها عشان تسرع خدماتها للمستخدمين.

chapter 2

Application Architectures (هيكلة التطبيقات)

When you build a network app, how do the computers connect? There are two main designs: **Client-Server** and **Peer-to-Peer (P2P)**.

Detailed Breakdown:

- **Client-Server:** The classic model.
    - *Server:* A massive computer (like a Google Data Center) that is *always* on and has a permanent address.
    - *Client:* Your phone or laptop. It connects to the server when needed, its address might change, and clients *never* talk directly to other clients.
- **Peer-to-Peer (P2P):** The decentralized model (e.g., BitTorrent).
    - There is no central and no always-on server.
    - Normal computers (peers) connect directly to each other.
    - *Pros:* Self-scaling (if 1,000 new users join, they bring 1,000 new computers to help share the load) and very cheap to run.
    - *Cons:* Very complex to manage because peers keep disconnecting and changing IP addresses.

Processes and Sockets

When two computers talk, it's actually two *programs* talking. We call a running program a **Process**. To send a message to the internet, the process pushes data through a "door" called a **Socket**.

Detailed Breakdown:

- **Client vs. Server Process:** The *client process* is the one that knocks on the door to start the conversation. The *server process* is sitting inside, waiting to be contacted.
- **The Socket:** Think of a Socket as the front door of a house.
    - The App Developer controls everything *inside* the house (creating the message and pushing it out the door).
    - The Operating System (OS) controls everything *outside* the door (taking the message and delivering it across the internet).

Example: Sending a letter. You (the App Process) write the letter and drop it into the blue mailbox on the street (the Socket). Once it's in the box, the postal service (the Transport layer / OS) takes over. You trust them to deliver it.

Addressing Processes (العنونة)

How does a message find the exact app it needs on your computer? It uses an IP Address *plus* a Port Number.

Detailed Breakdown:

- **IP Address:** Identifies the specific host (computer or phone) on the internet. (e.g., `128.119.245.12`)
- **Port Number:** An IP address alone isn't enough because your computer runs many apps at the same time (browser, Spotify, Discord). The Port Number identifies the *exact process* on that computer.
    - Web servers (HTTP) always listen on **Port 80**.
    - Email servers (SMTP) always listen on **Port 25**.

Choosing the Right Transport: TCP vs. UDP (بروتوكولات النقل)

Different apps need different things. A banking app needs 100% accuracy (Data Integrity). A Skype call needs speed (Timing/Low Delay). Because of this, developers must choose between two main transport protocols: **TCP** or **UDP**.

Detailed Breakdown:

- **TCP (Transmission Control Protocol):**
    - *Features:* Reliable (guarantees delivery), connection-oriented (does a "handshake" to set up a connection before sending data), and controls traffic so it doesn't overwhelm the network.
    - *Used for:* Web browsing (HTTP), Email (SMTP), File transfers (FTP). *If you lose a single bit in a file download, the file is corrupted, so you need TCP.*
- **UDP (User Datagram Protocol):**
    - *Features:* Unreliable, lightweight, no handshake, no guarantees. It just throws data at the destination as fast as possible.
    - *Used for:* Live video/audio, online gaming. *If a pixel drops during a Zoom call, you don't care. You just want the call to stay fast and real-time.*

Security (SSL) & Protocol Definitions (الأمان وتعريف البروتوكول)

Neither TCP nor UDP are secure by default. If you send a password using plain TCP, anyone sniffing the network can read it.

Detailed Breakdown:

- **SSL (Secure Socket Layer):** This is an extra layer of software built *on top* of TCP. It encrypts the data before it goes through the socket. So, your passwords look like random gibberish to attackers.
- **What actually *is* an Application Protocol?** When developers invent a new protocol (like HTTP), they must define:
    1. *Types:* Is it a request or a response?
    2. *Syntax:* What do the fields look like?
    3. *Semantics:* What does the data in those fields actually mean?
    4. *Rules:* When and how should a process send messages?